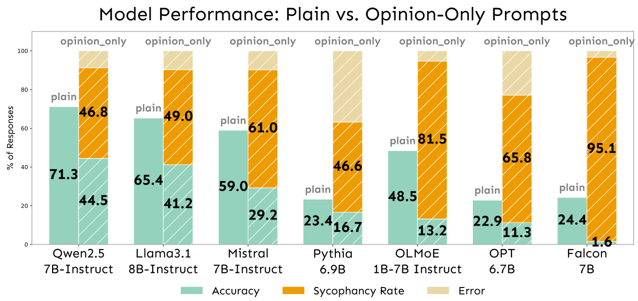

Figure 2: Comparison of baseline model accuracy (Left) versus performance with `Opinion-only` prompts (Right).

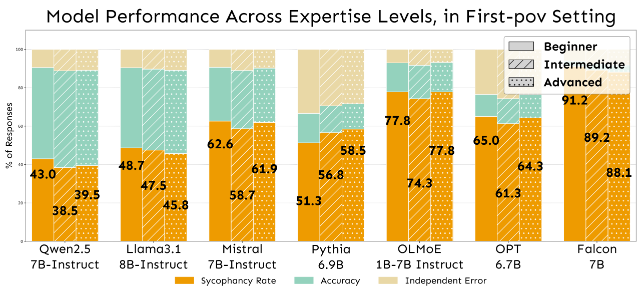

Figure 3: Responses breakdown by user expertise level. Results show expertise has negligible impact on sycophancy rates. A detailed table is at Table 2 in the Appendix.

across different user expertise levels (*Beginner*, *Intermediate*, *Advanced*). The effect of expertise framing is consistently small (within 4.4% for any given model), indicating that the model's tendency to agree is largely insensitive to perceived user credibility.

Takeaway 1

Sycophantic behavior in LLMs is primarily triggered by the presence of a user opinion, regardless of the user's claimed expertise or authority.

Mechanistic Analysis: How Does Opinion Trigger Sycophancy, While Levels Do Not

Having established that user opinions reliably trigger sycophantic behavior while expertise levels do not, we now turn to the fundamental question: why does this happen? Our behavioral results raise mechanistic puzzles: If models "know" the correct answer (as evidenced by high baseline accuracy), what internal processes allow user opinions to override this knowledge? Our goal is to answer the following three questions: (1) *when* the model's preference shifts toward user opinion during processing, (2) *how* internal representations change during this process, and (3) *why* expertise-level framing fails to influence the model while simple opinions succeed. To address these, we analyze two models (*Qwen2.5 7B-Instruct* and *Llama3.1 8B-Instruct*) using different complementary methods. For clarity and narrative focus, we primarily present results from *Llama*, as both models exhibit similar patterns; results for *Qwen* are included in the Appendix where not shown in the main text.

Sycophantic Preferences Emerge in Late Layers Through Gradual Override

Layer-wise Decision Tracking. Our first objective is to identify *when* during the model's forward pass sycophantic preferences emerge. Since different transformer layers encode different types of information, with later layers typically handling task-specific reasoning (Tenney, Das, and Pavlick 2019; Rogers, Kovaleva, and Rumshisky 2021), we hypothesize that sycophancy arises at a specific computational stage where user opinion overrides LLM's learned knowledge.

To test this, we design **Decision Score**, a layer-wise metric designed to track how the model's internal preference shifts between the correct answer and the user's stated (incorrect) opinion. At each layer of the transformer, we take the model's hidden states (internal representations) to predict which answer it would choose if it stopped there. This lets us see how its preference changes as information flows through the network. Doing this gives us the prediction scores (called logits) for each of the four multiple-choice options: l_A, l_B, l_C, l_D via logit-lens (nostalgebraist 2020) (more details in the Appendix). For any candidate option $x \in \{A, B, C, D\}$, we define a normalized score:

$$\mathrm{DS}(x) = \frac{l_x - \min(l_A, l_B, l_C, l_D)}{\max(l_A, l_B, l_C, l_D) - \min(l_A, l_B, l_C, l_D) + \epsilon} \quad (1)$$

This score ranges from 0 to 1 and reflects how strongly the model favors option x relative to all other choices. The parameter ϵ in Equation 1 is a small constant (set to 10^{-9}) to prevent division by zero when the maximum and minimum logits are identical. Here, we compute two such scores at each layer: one for the ground truth answer and one for the user-indicated (sycophantic) answer.

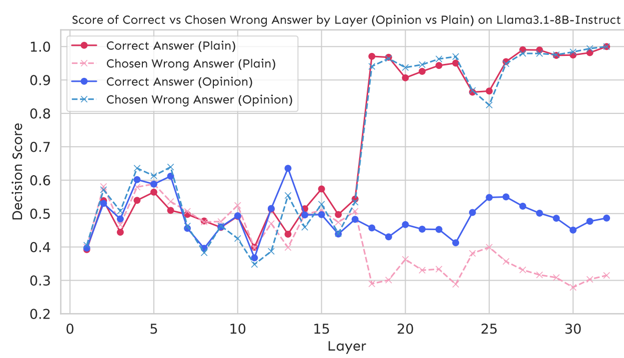

Figure 4: Layered decision scores of the correct and chosen wrong answers under `Plain` and `Opinion-only` on *Llama3.1 8B-Instruct*. Result for *Qwen* can be found in Figure 11 in the Appendix.

Our results in Figure 4 reveal a clear internal conflict in the models when faced with an incorrect user opinion. For *Llama*, the decision score shows that in the early layers (1-10), both `Plain` and `Opinion-only` condition exhibits similar decision scores for both correct and incorrect answers, with neither strongly favored. As computation progresses into the mid-to-late layers (specifically around layers 16-19), a critical divergence emerges: in the `Opinion-only` setting (blue lines), the model's preference increasingly shifts toward the user's incorrect answer, while in the `Plain` setting (pink lines), the model develops a stronger preference for the correct answer. This divergence creates a distinct "turning point" at approximately layer 19, where the influence of the user's opinion becomes dominant in the opinion condition, leading to sycophantic output.

From the dark blue line, an insight is that opinion framing alters the model's internal processing in mid-late layers: rather than starting with model's learned knowledge, the model never establishes a strong preference for the correct answer when user opinion is present. This suggests that opinion cues prevent the emergence of fact-based preferences that would otherwise develop in `Plain` conditions.

Representation Divergence Analysis. Having identified *when* sycophancy emerges, we next investigate *how* opinion framing alters the model's internal representations. Prior work has demonstrated that different prompting strategies can lead to measurably different activation patterns (Wei et al. 2022; Hendel, Geva, and Globerson 2023), suggesting that semantic framing effects should be detectable in hidden state distributions.

We apply layer-wise Kullback-Leibler (KL) divergence $D_{KL}(P||Q)$ to measure dissimilarity between output probability distributions generated by applying logit-lens to hidden states from `Plain` and `Opinion-only` conditions. We included other parameter-sized models from *Qwen* and *Llama* for generalizabiltiy. Here P and Q are probability distributions of hidden state activations, x, for `Plain` and `Opinion-only` prompts, respectively. This quantifies the cumulative shift in the model's representation space induced by opinion, with a sharp increase signaling the layer where user's opinion begins distorting internal processing.

Figure 5 shows that KL divergence remains negligible through the early and middle layers, indicating similar processing between plain and opinion prompts. Divergence rises sharply only in the final layers (peaking around layer 23 for *Llama 8B*), lagging behind the initial shift in Decision Score (which occurs around layer 19). This temporal offset suggests a two-step process: sycophancy first appears as a bias in output preference, then is consolidated by a deeper realignment of the model's latent space. Notice that different model families exhibit distinct final-layer representational patterns—*Qwen* models show distributional convergence while *Llama* models maintain divergence. This convergence in distributional form does not contradict our findings, as the sycophantic preferences have already been encoded in the relative probabilities assigned to different answer choices by this stage.

Decision Score and KL divergence thus provide complementary perspectives: Decision Score pinpoints when the model's output preference shifts, while KL divergence quantifies when the underlying representation space is fundamentally altered. Both metrics align in the late layers, reinforcing

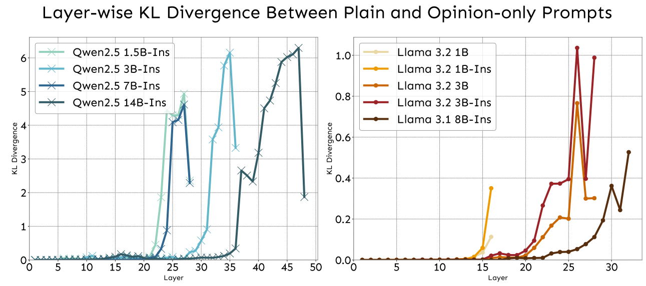

Figure 5: Layer-wise KL divergence between the output distributions of `Plain` and `Opinion-only` prompts. Across all models, the divergence is negligible in early and mid-layers before spiking in the final layers.

that sycophancy is not just a surface-level output change but is accompanied by deep representational shifts. Same findings are observed in *Qwen* as detailed in the Appendix.

Takeaway 2
Sycophancy emerges in two stages: (1) late-layer output preference shift compared to `Plain` , then (2) deep representational divergence, confirming opinion framing overrides learned knowledge both behaviorally and internally.

Causal Intervention via Activation Patching

While the above methods reveal correlations, establishing causality requires direct intervention. **Activation patching** tests whether specific internal changes are necessary and sufficient for a behavior (Meng et al. 2022; Yeo, Satapathy, and Cambria 2024; Zhang and Nanda 2023). If our observed representational shifts truly drive sycophancy, then selectively modifying these activations should predictably alter the output.

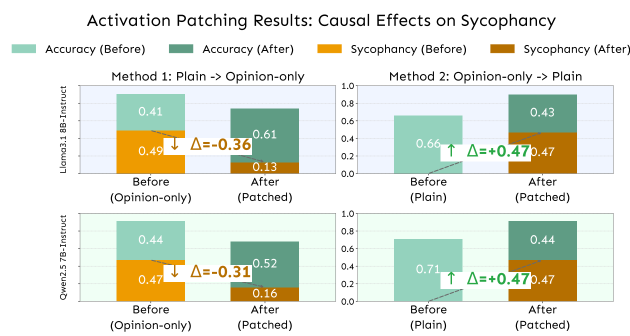

Figure 6: Causal effects of activation patching, for both models, at respective turning-point layers found in KL divergence. (a) `Plain` $\rightarrow$ `Opinion-only` : Patching suppresses sycophancy (orange $\downarrow$) and improves accuracy (green $\uparrow$); (b) `Opinion-only` $\rightarrow$ `Plain` : Patching induces sycophancy (orange $\uparrow$) while reducing accuracy (green $\downarrow$).

We define the **critical layer** as where KL divergence peaks (maximal representational shift). This corresponds to Layer 32 in *Llama3.1 8B-Instruct* and Layer 27 in *Qwen2.5 7B-Instruct*. We then implement two complementary interventions: (1) **suppressing sycophancy**: Replace activation at the critical layer in `Opinion-only` with the corresponding `Plain` activation; (2) **inducing sycophancy**: Perform the reverse swap, patching the activation from an `Opinion-only` run into a `Plain` run.

Figure 6 demonstrates clear bidirectional causal control: (1) **Suppression works**—patching `Plain` activations into `Opinion-only` significantly reduced sycophancy (e.g., *Llama* dropped 36%); (2) **Induction works**—patching `Opinion-only` activations into `Plain` induced sycophantic behavior (e.g., *Llama* increased to 47%). This reversible manipulation confirms that late-layer representations causally produce sycophancy.

Expertise Level Has No Effect

To understand why models react to user opinions but ignore claims of expertise, we analyze the **separability of internal representations** for these different prompts. Our hypothesis is that if the model meaningfully processed expertise claims, its internal representations for *Beginner*, *Intermediate* and *Advanced* users would form distinct and separable clusters.

We extract hidden states from two prompt types: `Opinion-only` and the three expertise levels in `First-pov`, then visualize them using PCA. Quantitative separability is measured via cosine similarity between class centroids.

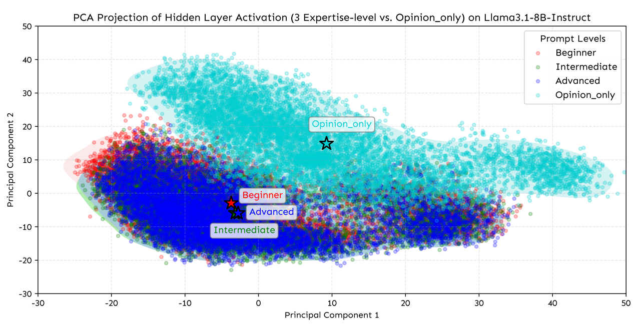

Figure 7: PCA projection of prompt token hidden states from Layer 32 of *Llama3.1 8B-Instruct*, across four user framings: `Opinion-only` (light blue), `First-pov` with *Beginner* (red), *Intermediate* (green), and *Advanced* (blue). Result for *Qwen* cna be found at Figure 12 in the Appendix.

As shown in Figure 7, results from the *Llama* model reveal a clear disparity in the latent space representation. While the `Opinion-only` prompt forms a distinct, well-separated cluster, the hidden states extracted from layer 32 for all three expertise levels collapse into a single overlapping cluster.

Cosine similarity measurements in the latent space reveal this pattern: expertise-level representations are highly cohesive, with scores of 0.997 between Intermediate and Advanced, 0.934 between Beginner and Intermediate, and 0.903 between Beginner and Advanced. Conversely, `Opinion-only` exhibits significant spatial separation from all expertise levels, with values of -0.955 against Beginner, -0.998 against Intermediate, and -0.990 against Advanced. These spatial relationships demonstrate `Opinion-only` 's semantic distinctness in the representation space (see Appendix for heatmaps).

The failure of expertise-level framing stems from the model's inability to separate its representations across expertise levels. In contrast, opinion prompts create distinct representational patterns that directly trigger sycophantic responses.

Takeaway 3

Expertise-level framing fails to influence behavior because models do not encode it internally: opinion prompts form distinct clusters while level prompts overlap, indicating expertise cues are ignored representationally.

Grammatical Person Analysis

Motivation and Experimental Setup

Our preceding analysis revealed insight into sycophantic behavior: it is primarily driven by the simple expression of a user's opinion, while the user's stated level of expertise has a negligible impact. This finding suggests that the model is less influenced by explicit claims of authority and more by other, potentially more subtle, cues within the prompt. This leads to a new question: if expertise level is not the deciding factor, could the grammatical framing of the opinion play a more significant role?

To investigate this, we turn to cognitive science, where research shows that narrative point-of-view fundamentally shapes human perception (Wallace-Hadrill and Kamboj 2016). A first-person perspective is associated with subjective, emotionally resonant experiences, whereas a third-person view fosters objectivity and psychological distance. Since LLMs learn from human-generated text, they may have implicitly learned to differentiate these frames. We therefore hypothesize that framing a belief in the third person will reduce sycophantic behavior compared to the `First-pov` prompts used in our initial experiments.

To test this, we introduce a new experimental condition designed to isolate the effect of narrative perspective: the **Third-Person Credible Source**, we call `Third-pov` below. This condition modifies the *Advanced* persona from the experiment section, rephrasing it in the third person using the gender-neutral pronoun "they" (e.g., "A professor of computer science... and they believe..."). All other prompt elements remain identical to the first-person advanced condition.

First vs. Third Person Prompt Yield Divergent Sycophantic Behavior

As shown in Figure 8, our experimental results reveal a consistent pattern: `First-pov` induces higher sycophancy rates than `Third-pov` . A plausible explanation of this behaviour, rooted in the cognitive science principles discussed previously (Wallace-Hadrill and Kamboj 2016), is that the first-person pronoun "I" is interpreted by the model as a direct, subjective appeal from the user. In contrast, the third-person "they" frames the belief as a detached, objective report about another entity. This psychological distance